

User Story

User Authentication & Authorization

The BanglaFactGuard system supports four types of users: **Anonymous User, Registered User, Expert, and Administrator**. Anonymous users can verify claims without creating an account, allowing quick access to the verification service. However, they cannot access submission history, personalized features, or notifications. Registered users can create an account by providing their name, email address, and password. After successful registration, users can log into the system using their credentials. If a user forgets their password, the system allows password recovery through a one time password sent to the registered email address.

The administrator is responsible for managing expert accounts. Experts are not allowed to register themselves; instead, the administrator creates expert accounts after verifying their professional credentials. Administrators can activate, deactivate, or update expert accounts whenever necessary. Both the website and browser extension share the same authentication system, allowing registered users to access their accounts seamlessly from either platform.

User Profile

Every registered user has a personal profile containing their basic information, including name, email address, profile details, and verification history. Users can view and update their personal information such as name, email, and contact details. They can also change their password securely through the system.

In addition, users can access their complete activity history, including all submitted claims, verification results, AI predictions, source verification outcomes, and final verdicts. This allows users to track the status and outcome of every fact-check they have performed.

Expert users have an additional profile containing their area of expertise, credibility score, completed reviews, and review history. Administrators can register, update, activate, and deactivate expert accounts, as well as register, modify, activate, deactivate, and delete verified news sources. They can also view and manage all registered users and experts, monitor account status, and perform other administrative actions whenever required.

Claim Submission & Verification

BanglaFactGuard provides three different verification services: **Multimodal Fact Verification, Source-Based Fact Verification**, and **Photo Card Verification**. Users can choose the appropriate verification method depending on the type of content they want to verify.

Multimodal Fact Verification

This verification method is designed to detect fake news using both textual and visual information. Users are required to provide the **news body text** and the corresponding **news image**, while the **news headline** is optional. The submitted textual and visual information is analyzed simultaneously using the multimodal ML/DL model to generate a preliminary prediction indicating whether the news is **Fake** or **Non-Fake**.

Source-Based Fact Verification

This verification method determines whether a news claim was actually published by a specific newspaper or media outlet. Users must provide the **news headline**, **news body text**, and select the **claimed news source** from the list of administrator-approved verified sources. The **publication date** is optional and may be supplied to improve evidence retrieval. The system searches the selected source for relevant articles, compares the submitted claim with the retrieved evidence, and returns one of the following verification results: **True**, **False**, **Partially True**, or **Not Found**.

Photo Card Verification

This verification method is intended for verifying news posters, social media photo cards, and screenshots containing textual information. Users are required to upload a **photo card or image**. They may also optionally provide the **headline**, **claimed news source**, and **publication date**. The system first extracts the embedded Bangla text using Optical Character Recognition (OCR) and presents the extracted text for confirmation or editing before verification begins. The extracted information is then compared with evidence retrieved from the selected source, and the final verification result is reported as **True**, **False**, **Partially True**, or **Not Found**.

For all verification methods, every submitted request is stored in the system. Before initiating a new verification process, the system checks whether the same or a highly similar claim has already been verified. If an existing verification result is found, the system reuses the previous result to reduce processing time and computational cost while informing the user that a similar claim has already been verified.

Anonymous users can use all three verification services without creating an account. However, they cannot access a permanent submission history, personalized notifications, or other account-specific features.

AI-Based Fact Verification

BanglaFactGuard combines automated artificial intelligence with human expertise to provide reliable fact verification. Depending on the selected verification type, the system performs different verification procedures.

For **Multimodal Fact Verification**, the submitted textual and visual content is analyzed simultaneously by a multimodal Machine Learning/Deep Learning model. The model produces an initial prediction indicating whether the submitted news is **Fake** or **Non-Fake**. This prediction serves only as a preliminary assessment and is not considered the final verification result.

For **Source-Based Fact Verification**, the system retrieves related articles from the selected verified news source and performs similarity analysis between the submitted claim and the retrieved articles. Based on the evidence, the system classifies the claim as **True**, **False**, **Partially True**, or **Not Found**.

For **Photo Card Verification**, the system first extracts textual information from the uploaded image using OCR. The extracted text, together with any optional metadata provided by the user, is compared against articles retrieved from the selected verified news source. The verification result is then categorized as **True**, **False**, **Partially True**, or **Not Found**.

The AI-generated prediction and source-based verification are treated as preliminary assessments. To improve reliability and minimize incorrect decisions, expert reviewers examine the submitted content, AI prediction, retrieved evidence, and supporting information before providing their own verdict. The final verification result is determined using the credibility-weighted expert voting mechanism and is presented together with the supporting evidence and verification details.

Browser Extension

The BanglaFactGuard Browser Extension enables users to verify online news content directly from their web browser without opening the main website. Registered users can log into the extension using the same credentials as the web application, allowing all verification requests performed through the extension to be synchronized automatically with their account and appear in their submission history.

The browser extension provides access to all three verification services: **Multimodal Fact Verification**, **Source-Based Fact Verification**, and **Photo Card Verification**, enabling users to verify different types of content while browsing online.

For **Multimodal Fact Verification**, users can manually input the required **news body text** and upload the corresponding **news image**. Additionally, the extension allows users to **select text directly from a webpage and capture a screenshot automatically**, which can be used as the required image input. Users may also upload an image manually from their device. The **news headline** is optional. The submitted textual and visual information is then processed using the multimodal ML/DL model to generate a preliminary prediction indicating whether the news is **Fake** or **Non-Fake**.

For **Source-Based Fact Verification**, users can select or highlight news content from a webpage, and the extension will automatically extract the **news headline** and **news body text**. Users must also select the **claimed news source** from the administrator-maintained list of verified sources. The **publication date** is optional and may be provided if available. The system then processes the submitted information and performs verification based on the provided inputs.

For **Photo Card Verification**, users can upload a **photo card, screenshot, or image** directly through the extension. The extension also allows users to capture a **screenshot of selected content from a webpage** or upload an image manually from their device. The system extracts the embedded Bangla text using Optical Character Recognition (OCR) and presents it to the user for confirmation or editing before verification begins. Users may also optionally provide the **headline, claimed news source, and publication date** to improve verification accuracy.

After a verification request is completed, the extension displays a concise verification summary containing the verification type, final verification result, and a brief evidence summary. It also provides a link to the complete verification report on the BanglaFactGuard website. Users can open the detailed report to view the full verification process, supporting evidence, references, expert review summary, and final verification verdict.

Expert Review

To improve the reliability of automated verification, BanglaFactGuard incorporates expert-assisted validation. Expert users can access a review dashboard containing pending submissions awaiting manual verification. For each submission, experts can review the submitted claim, uploaded image, OCR-extracted text, AI prediction, source verification results, and supporting evidence before submitting their own verdict and written justification.

Each expert's vote contributes to the final decision according to their credibility score. After a verification process is finalized, the system automatically updates each expert's credibility score based on the consistency of their reviews with the final verdict, ensuring that experienced and accurate experts have greater influence in future verifications.

Verified Source Management

The administrator manages the repository of trusted news sources used for source-based verification. Administrators can register new verified news sources by providing detailed source information such as the source name, English display name, website URL, search URL, RSS feed (optional), article extraction selectors, article URL patterns, language, and trust status.

Administrators can edit, enable, disable, or remove verified sources whenever necessary. These verified sources are automatically displayed in both the website and browser extension as selectable options during claim submission.

Fact Explorer

The **Fact Explorer** serves as a public repository of all published fact-checks available in the BanglaFactGuard system. It allows both registered users and anonymous visitors to browse previously verified claims without submitting a new verification request. The repository includes fact-checks generated through **AI Verification**, **Source Verification**, and **Photo Card Verification**, providing a centralized archive of verified information.

Users can search and filter fact-checks using keywords, claim titles, verification types, publication dates, or claimed news sources. Each fact-check is presented as a card displaying a brief summary, verification type, final verdict, publication date, and thumbnail image (if available). Selecting a fact-check redirects the user to a dedicated detail page containing the complete verification report, including the original claim, verification

methodology, AI prediction (if applicable), source verification results, supporting evidence, references, confidence score, related media, expert review summary, and the final verification verdict.

The Fact Explorer also includes a public analytics section that provides an overview of platform activity and misinformation trends. Visitors can view statistics such as the total number of verified claims, fake claims, non-fake claims, uncertain claims, pending reviews, verification types, most frequently claimed news sources, and the average verification time.

In this page there should also show the available experts profile. all type user can search and see an experts details

Notification System

The system automatically notifies registered users whenever the status of one of their submitted claims changes. Users receive notifications when AI analysis is completed, when the source based verification is completed for their claim, when expert review has been finalized, or when the final verification report becomes available. Notifications are accessible through the website and may also be displayed within the browser extension for logged-in users. Experts receive notifications whenever new submissions become available for review.

Administration & System Monitoring

The administrator is responsible for managing the core operational components of the BanglaFactGuard platform while retaining all the capabilities available to a registered user. In addition to submitting claims, viewing verification results, and accessing the Fact Explorer, administrators can manage expert accounts and verified news sources.

The administrator can register new expert accounts by providing the necessary professional information, update expert profiles, activate or deactivate expert accounts, and control their participation in the expert review process.

The administrator is also responsible for maintaining the repository of verified news sources used for source-based verification. They can register new verified news sources by providing the required source information, update existing source configurations, activate or deactivate sources, and permanently delete sources that are no longer required. These verified sources are automatically available as selectable options in both the website and browser extension during claim submission.

Additionally, the administrator can define and update the **expert credibility scoring mechanism** used in the final verification process. This mechanism allows the system to assign different weights to expert votes based on their historical accuracy. For example, the administrator may configure rules such as: if an expert has achieved a credibility accuracy of 90% or above, their vote will be given higher weight (e.g., counted as double), whereas experts with lower credibility scores will have reduced influence in the final decision. This configurable weighting system ensures that more reliable experts have a greater impact on the final verdict while maintaining fairness and adaptability in the expert evaluation process.
